

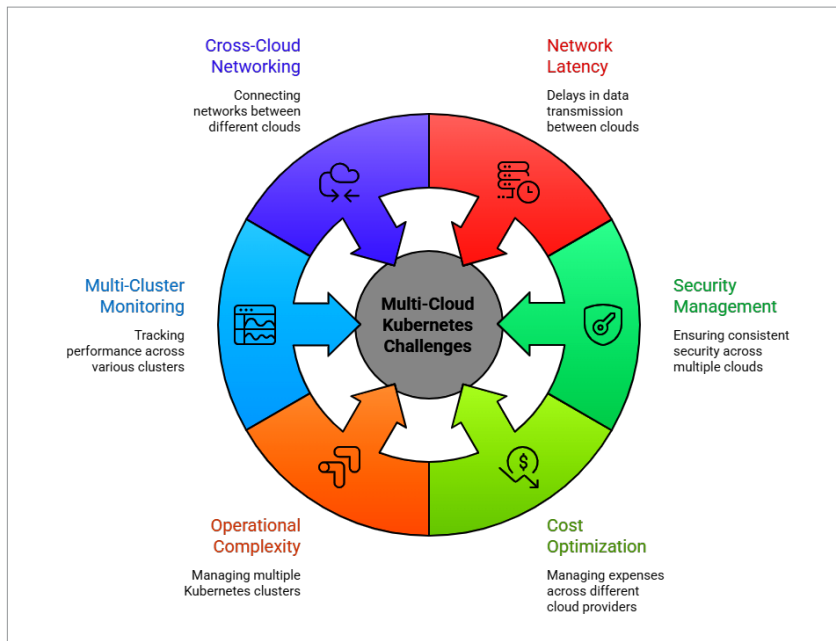


Figure 4: Challenges in managing multi-cloud Kubernetes

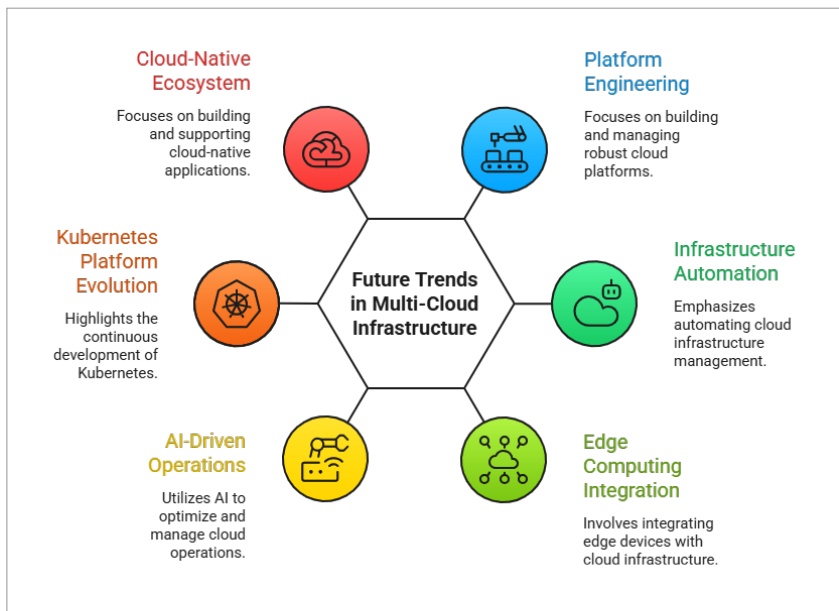


Figure 5: The future of multi-cloud infrastructure

Operational complexity: Effective automation and monitoring systems are required for each cluster to avoid operational issues.

Organisations that utilise multi-cloud architectures must establish strong governance frameworks and standardised infrastructure templates for all their clusters.

The future of multi-cloud infrastructure

The following trends will impact the future of multi-cloud infrastructure.

Platform engineering: Enterprise internal developer platforms are becoming more commonplace.

Infrastructure automation: Tools that provide automatic lifecycle management of clusters are rapidly advancing and will likely become standard.

Integration of edge computing: Use of Kubernetes for managing workloads across edge facilities is becoming more common.

AI-driven infrastructure operations: Infrastructure optimisation and anomaly detection can now be achieved using machine learning algorithms.

The application of Kubernetes as a base platform has led to it being the cornerstone for managing container workloads that are distributed across many different cloud providers. By combining Kubernetes with other open source control plane technologies like GitOps, Cluster API, and CrossPlane, Kubernetes can provide more than just the orchestration of applications. It can also serve as an integrated platform for managing all aspects of an organisation's infrastructure.

As the ecosystem matures, Kubernetes and open source control planes will most likely play key roles in driving the evolution of cloud native infrastructure platforms. **END** 

Reference

- <https://www.flexera.com/blog/finops/cloud-computing-trends-flexera-2024-state-of-the-cloud-report/>

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